



# ROBERT VOKÁČ, BSc

## Software Engineer – C++ Systems & Cross-Platform Development

Prague, Czech Republic · +420 704 780 275 · [robertvokac@robertvokac.com](mailto:robertvokac@robertvokac.com)

**GitHub:** [github.com/openeggbert](https://github.com/openeggbert) · [github.com/robertvokac](https://github.com/robertvokac)

**Web:** [robertvokac.com](https://robertvokac.com) · **LinkedIn:** [linkedin.com/in/robert-vokac](https://linkedin.com/in/robert-vokac)

## PROFILE

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Software Engineer with substantial independent work in **modern C++ systems programming**.

I build **3D tooling** (a 3D scene editor and a procedural world generator), **cross-platform frameworks**, compatibility layers, runtime systems, and platform abstractions using C++23, **SDL3**, **OpenGL**, bgfx, and CMake. My work focuses on portable architecture, API compatibility, rendering/input/audio systems, and application portability across Windows, Linux, WebAssembly, and Android. My public portfolio spans **≈495,500 lines of C++** across the projects below.\*

Seeking roles in systems programming, framework engineering, tooling, platform abstraction, or cross-platform software development.

## RELEVANT C++ ENGINEERING EXPERIENCE

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### Independent C++ Systems Developer – OpenEggbert

Open-source engineering work · 2024 – present

Designed and built modern C++ 3D tooling, framework, and compatibility-layer projects focused on portability, API behavior reproduction, and platform abstraction.

Main technologies: **C++20/23**, **SDL3**, **OpenGL**, **bgfx**, **CMake**, **Windows**, **Linux**, **WebAssembly**, **Android**

#### Mesh Craft — 3D Scene Editor (flagship project) ≈29.4k LOC

**GitHub:** [github.com/openeggbert/mesh-craft](https://github.com/openeggbert/mesh-craft) · **Web:** [meshcraft3d.com](https://meshcraft3d.com) · [meshcraft.openeggbert.com](https://meshcraft.openeggbert.com)

C++23 desktop 3D scene editor for the XML-based MC3 format — an editable, human-readable scene description compiled to standard runtime formats.

- Designed the `.mc3.xml` scene format: primitives, hierarchical groups, PBR materials, prefabs, layers
- Implemented CSG boolean operations (union/difference/intersection via Manifold) and extrude-along-path
- Built keyframe animation (position, rotation, scale, visibility, material channels) with a timeline editor
- Created glTF/GLB and binary MCB exporters with CLI tools (`mc3togltf`, `mc3tomcb`)
- Built the Dear ImGui editor (orbit camera, gizmos, undo/redo, autosave) on the CNA runtime (SDL3/OpenGL ES 3)

## Mesh World — Procedural 3D World Generator ≈16.9k LOC

**GitHub:** [github.com/openeggbert/mesh-world](https://github.com/openeggbert/mesh-world) · **Web:** [meshworld3d.com](https://meshworld3d.com) · [meshworld.openeggbert.com](https://meshworld.openeggbert.com)

Procedural 3D world/city generator built on the Mesh Craft ecosystem, with a real-time exploration app.

- Implemented 20 C++ chunk generators and 11 Lua object generators emitting MC3 scenes
- Embedded a Lua 5.4 scripting sandbox (sol2) for content modding with auto-discovered generators
- Built a planetary map subsystem (quadtree LOD 0-18, 52-biome classification, hydrology and mountain-range terrain algorithms)
- Designed SQLite content packs (taxonomy/material registries, generator bundles) and an MC3 validation pipeline
- Built MeshWorldApp, a standalone SDL3/OpenGL explorer streaming chunks around the player
- 1176 tests passing, zero warnings, CI on every push

## CNA — Cross-Platform C++ Framework (XNA-style API) ≈181.2k LOC

**GitHub:** [github.com/openeggbert/cna](https://github.com/openeggbert/cna) · **Web:** [libcna.com](https://libcna.com) · [cna.openeggbert.com](https://cna.openeggbert.com) · **Demo:** [speedyblupi.com/SpeedyBlupi2013](https://speedyblupi.com/SpeedyBlupi2013) (WebAssembly)

Cross-platform C++ framework implementing an XNA-style programming model over SDL3, validated by a working C# XNA game port running natively on desktop, Android, and WebAssembly.

- Designed a cross-platform C++ framework layer over SDL3
- Built application loop, input, rendering, audio, and resource systems
- Grew the pluggable rendering backend layer to 12 backends: SDL\_Renderer, OpenGL (EasyGL), Vulkan, and bgfx, plus WebGPU, native Direct3D 9/11/12 (Wine/DXVK-verified), SDL\_GPU (Vulkan-driven), HTML Canvas, an ASCII glyph-grid backend, and a DirectDraw-shaped backend (DX3) fronting the free-direct project
- Ported a C# / XNA application to native C++, replacing .NET, MonoGame, and original XNA dependencies
- Produced stable builds from one codebase for Windows, Linux, WebAssembly, and Android

## CNA Samples — XNA Sample Collection Ported to C++ ≈48.0k LOC

**GitHub:** [github.com/openeggbert/cna-samples](https://github.com/openeggbert/cna-samples) · **Demo:** web demos coming soon

C++ ports of the official Microsoft XNA Game Studio 4.0 sample collection, running on CNA — a continuously growing compatibility test bed proving the framework's API coverage on real code.

## CNA Craft — Voxel World Prototype on CNA ≈6.3k LOC

**GitHub:** [github.com/openeggbert/cna-craft](https://github.com/openeggbert/cna-craft) · **Status:** in development

Minecraft-like first-person voxel-world prototype built directly on the CNA framework, a faithful C++ port of fogleman/Craft: unbounded chunk-streamed block terrain with baked ambient occlusion, a 54-block roster, world-editing commands, signs, SQLite delta-based world persistence, and multiplayer against its own server. Runs on CNA's OpenGL (EasyGL), Vulkan, and bgfx backends, and is playable in the browser via WebAssembly/WebGL2.

## Sharp Runtime — C#/I.NET Runtime Subset in Native C++ ≈70.7k LOC

**GitHub:** [github.com/openeggbert/sharp-runtime](https://github.com/openeggbert/sharp-runtime) · **Web:** [sharpruntime.openeggbert.com](https://sharpruntime.openeggbert.com)

Reimplementation of a pragmatic subset of the .NET runtime (System.\* namespaces) in idiomatic modern C++: exceptions, events/delegates, collections, string/number/date formatting, and IO. Foundation layer for CNA, Mesh Craft, Mesh World, and the game ports.

## Free Direct — DirectX 3 (2D) Compatibility Layer ≈4.3k LOC

**GitHub:** [github.com/openeggbert/free-direct](https://github.com/openeggbert/free-direct) · **Web:** [freedirect.openeggbert.com](https://freedirect.openeggbert.com) · **Demos:** [speedyblupi.com/SpeedyEggbert2](https://speedyblupi.com/SpeedyEggbert2) · [speedyblupi.com/PlanetBlupi](https://speedyblupi.com/PlanetBlupi)

Modern C++ reimplementation of a game-driven DirectDraw / DirectSound subset over SDL3.

- Created DirectDraw-style 2D rendering abstractions: CPU surfaces, blitting, palettes, color keys, locking, presentation
- Reproduced selected DirectX 3 behavior without obsolete SDK dependencies

## Free API — Win32 Compatibility Layer ≈4.5k LOC

**GitHub:** [github.com/openeggbert/free-api](https://github.com/openeggbert/free-api) · **Web:** [freeapi.openeggbert.com](https://freeapi.openeggbert.com)

Minimal Win32 API (circa 1998) compatibility layer on SDL3 — windowing, message loop, input translation, multimedia timers, and MIDI/MCI music — enabling legacy Windows games to run on Linux, macOS, Android, and the web. Works together with Free Direct as the system/platform side of legacy application compatibility.

## Free Eggbert — Speedy Eggbert 2 Reconstruction ≈28.1k LOC

**GitHub:** [github.com/openeggbert/free-eggbert](https://github.com/openeggbert/free-eggbert) · **Web:** [freeeggbert.openeggbert.com](https://freeeggbert.openeggbert.com) · **Demo:** [speedyblupi.com/SpeedyEggbert2](https://speedyblupi.com/SpeedyEggbert2) (WebAssembly, partial)

Game-preservation project: a buildable reconstruction of the 1998–2001 game Speedy Eggbert 2, reverse-engineered from original binaries (Ghidra, IDA, ILSpy) and made portable beyond Windows/DirectX via Free API + Free Direct, including an Emscripten web build with persistent saves.

## Mobile Eggbert — C++ Port of Speedy Blupi (2013) ≈20.5k LOC

**GitHub:** [github.com/openeggbert/mobile-eggbert](https://github.com/openeggbert/mobile-eggbert) · **Web:** [mobileeggbert.openeggbert.com](https://mobileeggbert.openeggbert.com) · **Demo:** [speedyblupi.com/SpeedyBlupi2013](https://speedyblupi.com/SpeedyBlupi2013) (WebAssembly, fully playable)

C++ port of the 2013 Windows Phone XNA game Speedy Blupi, running on CNA. Migration chain: C# XNA 4.0 → ILSpy decompilation → MonoGame → native C++/CNA. Playable on desktop and in the browser (IndexedDB save persistence); the reference implementation for Galaxy Eggbert.

## Galaxy Eggbert — 3D Game on the CNA Stack ≈17.3k LOC

**GitHub:** [github.com/openeggbert/galaxy-eggbert](https://github.com/openeggbert/galaxy-eggbert) · **Web:** [galaxyeggbert.openeggbert.com](https://galaxyeggbert.openeggbert.com)

3D remake of Speedy Blupi / Mobile Eggbert validating the full CNA + Easy3D engine stack; targets Linux, Windows, WebAssembly, and Android from a single C++ codebase.

## easy-gl — OpenGL / OpenGL ES RAIL Wrapper ≈3.6k LOC

**GitHub:** [github.com/openeggbert/easy-gl](https://github.com/openeggbert/easy-gl) · **Web:** [easygl.openeggbert.com](https://easygl.openeggbert.com)

Toolkit-independent C++20 RAIL wrapper over OpenGL / OpenGL ES — shaders, buffers, VAOs, textures, framebuffers. The host app owns the window and GL context; used by CNA as its OpenGL rendering backend.

## easy-3d — 3D Helper Library for CNA ≈1.0k LOC

**GitHub:** [github.com/openeggbert/easy-3d](https://github.com/openeggbert/easy-3d) · **Web:** [easy3d.openeggbert.com](https://easy3d.openeggbert.com)

Small C++23 helper library beside CNA: cameras (orbit/follow), texture atlas, billboard and cube batching, and debug draw — deliberately a library, not an engine. First consumer: Galaxy Eggbert.

## meta-gl — Low-Level Type-Safe OpenGL Wrapper ≈8.3k LOC

**GitHub:** [github.com/openeggbert/meta-gl](https://github.com/openeggbert/meta-gl) · **Web:** [metagl.openeggbert.com](https://metagl.openeggbert.com)

Type-safe C++23 wrapper for OpenGL ES 2.0+ / WebGL: runtime function loading via a host-supplied GetProcAddress, enum class wrappers for GL constants, 358 thin wrapper functions. Foundation layer beneath easy-gl; Emscripten/WebGL support with context-loss handling.

## Mobile Eggbert Legacy — C#/MonoGame Preservation Archive

**GitHub:** [github.com/openeggbert/mobile-eggbert-legacy](https://github.com/openeggbert/mobile-eggbert-legacy) · **Stack:** C#, XNA 4.0, MonoGame

Preserved C#/MonoGame stage of the Mobile Eggbert migration: ILSpy-decompiled Windows Phone XNA sources with related repositories merged via git subtree.

## Mobile Eggbert LibGDX — Java Port

**GitHub:** [github.com/openeggbert/mobile-eggbert-libgdx](https://github.com/openeggbert/mobile-eggbert-libgdx) · **Stack:** Java, LibGDX, Gradle

Java/LibGDX port of Speedy Blupi (Mobile Eggbert) with a small XNA/.NET compatibility bridge; desktop (LWJGL3), Android, and HTML targets.

## Sprite Utils — Sprite Utilities and Assets ≈2.2k LOC

**GitHub:** [github.com/openeggbert/sprite-utils](https://github.com/openeggbert/sprite-utils)

Small C++23 sprite utility library and assets (number spritesheets, web component) supporting the game projects.

## YouTube Frontend — Static Index Generator for ArchiveBox ≈2.3k LOC

**GitHub:** [github.com/openeggbert/youtube-frontend](https://github.com/openeggbert/youtube-frontend) · **Web:** [youtube.openeggbert.com](https://youtube.openeggbert.com)

C++23 tool that generates static HTML index pages for ArchiveBox video archives, using OpenCV, FFmpeg libraries, and libcurl.

## Personal Projects – robertvokac

### Hive — Modular Backend Platform ≈45.0k LOC

**GitHub:** [github.com/robertvokac/hive](https://github.com/robertvokac/hive) · **Web:** [hive.robertvokac.com](https://hive.robertvokac.com)

Metadata-driven C++23 backend platform demonstrating large-scale application architecture, extensibility, and backend systems design.

- Designed a plugin-based architecture with dependency-aware startup and migrations
- Built a custom model/ORM layer with metadata-driven entity definitions
- Implemented auto-generated REST APIs from shared model metadata
- Developed scheduler, trigger, validation, and automation pipelines
- Added authentication, token management, operational logging, and migration integrity mechanisms
- Created schema-driven frontend generation backed by shared backend metadata

### bit-backup — Bit Rot Detection Tool ≈3.1k LOC

**GitHub:** [github.com/robertvokac/bit-backup](https://github.com/robertvokac/bit-backup) · **Stack:** C++23, SQLite, OpenSSL

Command-line tool that detects silent data corruption (bit rot) by storing and verifying SHA-512 file checksums in SQLite — parallel hashing, ignore patterns, rotating scrub verification, and CSV reporting for cron integration.

### Lexicon — Desktop Knowledge Dictionary ≈2.8k LOC

**GitHub:** [github.com/robertvokac/lexicon](https://github.com/robertvokac/lexicon) · **Web:** [lexicon.robertvokac.com](https://lexicon.robertvokac.com) · **Stack:** C++23, Qt 6 Widgets, SQLite

Desktop knowledge dictionary for structured technical notes: typed term relationships with backlinks, Markdown content with live preview, full-text search, and light/dark themes.

\* Code-line counts measured with cloc (July 2026): C++ sources and headers (.cpp/.hpp/.h), src/ and include/ directories only, excluding tests, vendored, and third-party code.

## PRODUCTION SOFTWARE EXPERIENCE

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### Software Engineer – Assist

2017 – present

Production software engineering role focused on backend systems, SQL, legacy codebases, and long-term maintenance.

- Maintained and improved production Java applications and SQL databases
- Diagnosed and resolved production incidents in large legacy applications
- Performed root-cause analysis, debugging, and long-term stability improvements

**Technologies:** Java, SQL, Git, Bash

## CORE SKILLS

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**Languages:** C++20/23, SQL, Java

**C++ / Systems:** STL, RAII, smart pointers, resource ownership, modular architecture

**Cross-platform:** Windows, Linux, WebAssembly, Android

**Graphics / Platform:** SDL3, OpenGL, bgfx, Dear ImGui, DirectDraw-style rendering, platform abstraction

**3D / Tooling:** scene editing, CSG, glTF/GLB pipelines, procedural generation, Lua scripting (sol2)

**Build / Tools:** CMake, Git, Bash, CLion, Visual Studio

**Backend / Data:** SQL, SQLite, REST API design, ORM concepts

**Core areas:** compatibility layers, legacy behavior reproduction, rendering/input/audio systems

## EDUCATION

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**BSc in Computer Science**

Czech University of Life Sciences Prague

2014 – 2017

## LANGUAGES

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**Czech:** Native

**English:** B2